

Exam Date 25 Jul 2022
Exam Slot 2

Question ID:156941
Topic Name: Mathematics-Section A

Question: For $z \in \mathbb{C}$ if the minimum value of $\left(\left|z-3\sqrt{2}\right|+\left|z-p\sqrt{2}i\right|\right)$ is $5\sqrt{2}$, then a value of p is _____.

A 3

B $\frac{7}{2}$

C 4

D $\frac{9}{2}$

Answer Given By Candidate:**C**

Question ID:156942
Topic Name: Mathematics-Section A

Question: The number of real values of λ, such that the system of linear equations

$2x-3y+5z=9$

$x+3y-z=-18$

$3x-y+(\lambda^2-|\lambda|)z=16$

has no solutions, is

A 0

B 1

C 2

D 4

Answer Given By Candidate:**C**

Question ID:156943
Topic Name: Mathematics-Section A
Question: The number of bijective functions $f : \{1, 3, 5, 7, \ldots, 99\} \to \{2, 4, 6, 8, \ldots, 100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \ldots \geq f(99)$, is _____

A ${}^{50}P_{17}$

B ${}^{50}P_{33}$

C $33! \times 17!$

D $\frac{50!}{2}$

Answer Given By Candidate:**B**

Question ID:156944
Topic Name: Mathematics-Section A
Question: The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is

A 1

B 4

C 6

D 8

Answer Given By Candidate:**Not Attempted**

Question ID:156945
Topic Name: Mathematics-Section A
Question: The sum $\sum_{n=1}^{21} \frac{3}{(4n-1)(4n+3)}$ is equal to

A $\frac{7}{87}$

B $\frac{7}{29}$

C $\frac{14}{87}$

D $\frac{21}{29}$

Answer Given By Candidate:**B**

Question ID:156946
Topic Name: Mathematics-Section A
Question: $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ is equal to

A 14

B 7

C $14\sqrt{2}$

D $7\sqrt{2}$

Answer Given By Candidate:**A**

Question ID:156947
Topic Name: Mathematics-Section A
Question: $\lim\limits_{n\to\infty} \dfrac{1}{2^n}\left(\dfrac{1}{\sqrt{1-\frac{1}{2^n}}}+\dfrac{1}{\sqrt{1-\frac{2}{2^n}}}+\dfrac{1}{\sqrt{1-\frac{3}{2^n}}}+....+\dfrac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ is equal to

A $\frac{1}{2}$

B 1

C 2

D -2

Answer Given By Candidate:**Not Attempted**

Question ID:156948
Topic Name: Mathematics-Section A
Question: If A and B are two events such that $P(A)=\frac{1}{3}, P(B)=\frac{1}{5}$ and $P(A\cup B)=\frac{1}{2}$, then $P(A\,|\,B')+P(B\,|\,A')$ is equal to

A $\frac{3}{4}$

B $\frac{5}{8}$

C $\frac{5}{4}$

D $\frac{7}{8}$

Answer Given By Candidate:**B**

Question ID:156949
Topic Name: Mathematics-Section A

Question: Let $[t]$ denote the greatest integer less than or equal to t. Then the value of the integral $\int_{-3}^{101}\left([\sin(\pi x)]+e^{[\cos(2\pi x)]}\right)dx$ is equal to

A $\frac{52(1-e)}{e}$

B $\frac{52}{e}$

C $\frac{52(2+e)}{e}$

D $\frac{104}{e}$

Answer Given By Candidate:**Not Attempted**

Question ID:1569410
Topic Name: Mathematics-Section A

Question: Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then $100(\alpha + \beta)$ is equal to

A -14

B 42

C -22

D 14

Answer Given By Candidate:**Not Attempted**

Question ID:1569411
Topic Name: Mathematics-Section A

Question: Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to

A $2\log_e 2$

B 4

C 1

D $4\log_e 2$

Answer Given By Candidate:**B**

Question ID:1569412
Topic Name: Mathematics-Section A

Question: If the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the line $\frac{x}{7}+\frac{y}{2\sqrt{6}}=1$ on the x-axis and the line $\frac{x}{7}-\frac{y}{2\sqrt{6}}=1$ on the y-axis, then the eccentricity of the ellipse is

A $\frac{5}{7}$

B $\frac{2\sqrt{6}}{7}$

C $\frac{3}{7}$

D $\frac{2\sqrt{5}}{7}$

Answer Given By Candidate:**A**

Question ID:1569413
Topic Name: Mathematics-Section A

Question: The tangents at the points $A(1, 3)$ and $B(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is:

A 4

B 6

C 7

D 8

Answer Given By Candidate:**D**

Question ID:1569414
Topic Name: Mathematics-Section A

Question: Let the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is :

A $\frac{32}{9}$

B $\frac{18}{5}$

C $\frac{27}{4}$

D $\frac{27}{10}$

Answer Given By Candidate:**Not Attempted**

Question ID:1569415
Topic Name: Mathematics-Section A

Question: A plane E is perpendicular to the two planes $2x-2y+z=0$ and $x-y+2z=4$, and passes through the point $P(1,-1,1)$. If the distance of the plane E from the point $Q(a, a, 2)$ is $3\sqrt{2}$, then $(PQ)^2$ is equal to

A 9

B 12

C 21

D 33

Answer Given By Candidate:**Not Attempted**

Question ID:1569416
Topic Name: Mathematics-Section A

Question: The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is

A $2\sqrt{29}$

B 1

C $\sqrt{\frac{37}{29}}$

D $\frac{\sqrt{29}}{2}$

Answer Given By Candidate:**A**

Question ID:1569417
Topic Name: Mathematics-Section A

Question: Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is :

A $\frac{2}{\sqrt{21}}$

B $2\sqrt{\frac{3}{7}}$

C $\frac{2}{3}\sqrt{\frac{7}{3}}$

D $\frac{2}{3}$

Answer Given By Candidate:**A**

Question ID:1569418
Topic Name: Mathematics-Section A

Question: If the mean deviation about median for the numbers 3, 5, 7, 2k, 12, 16, 21, 24, arranged in the ascending order, is 6 then the median is

A 11.5

B 10.5

C 12

D 11

Answer Given By Candidate:**Not Attempted**

Question ID:1569419
Topic Name: Mathematics-Section A

Question: $2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to :

A $\frac{3}{16}$

B $\frac{1}{16}$

C $\frac{1}{32}$

D $\frac{9}{32}$

Answer Given By Candidate:**Not Attempted**

Question ID:1569420
Topic Name: Mathematics-Section A

Question:
Consider the following statements :
P : Ramu is intelligent.
Q : Ramu is rich.
R : Ramu is not honest.

The negation of the statement "Ramu is intelligent and honest if and only if Ramu is not rich" can be expressed as :

A $((P \wedge (\sim R)) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee R))$

B $((P \wedge R) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$

C $((P \wedge R) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$

D $((P \wedge (\sim R)) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee R))$

Answer Given By Candidate:**A**

Question ID:1569421
Topic Name: Mathematics-Section B

Question:
Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Define $B = \{T \subseteq A : \text{either } 1 \notin T \text{ or } 2 \in T\}$ and $C = \{T \subseteq A : T \text{ the sum of all the elements of } T \text{ is a prime number}\}$. Then the number of elements in the set $B \cup C$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569422
Topic Name: Mathematics-Section B

Question:
Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p$, $p \neq 0$, and $f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and $fofofof(x) = 0$ have a common real root, then $f(-3)$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569423
Topic Name: Mathematics-Section B

Let $A=\begin{bmatrix}1 & a & a\\ 0 & 1 & b\\ 0 & 0 & 1\end{bmatrix}$, $a,b\in\mathbb{R}$. If for some $n\in\mathbb{N}$, $A^n=\begin{bmatrix}1 & 48 & 2160\\ 0 & 1 & 96\\ 0 & 0 & 1\end{bmatrix}$

Question: then $n+a+b$ is equal to _____.

Answer Given By Candidate:**68**

Question ID:1569424
Topic Name: Mathematics-Section B

The sum of the maximum and minimum values of the function $f(x)=|5x-7|+[x^2+2x]$ in the interval $\left[\frac{5}{4},2\right]$, where $[t]$ is the greatest integer $\le t$, is _____.

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:1569425
Topic Name: Mathematics-Section B

Let $y=y(x)$ be the solution of the differential equation

$$\frac{dy}{dx}=\frac{4y^3+2yx^2}{3xy^2+x^3},\ y(1)=1.$$

If for some $n\in\mathbb{N}$, $y(2)\in[n-1,n)$, then n is equal to _____.

Question:

Answer Given By Candidate:**3**

Question ID:1569426
Topic Name: Mathematics-Section B

Let f be a twice differentiable function on $\mathbb{R}$. If $f'(0)=4$ and $f(x)+\int_0^x(x-t)f'(t)\,dt=\left(e^{2x}+e^{-2x}\right)\cos 2x+\frac{2}{a}x$, then $(2a+1)^5\,a^2$ is equal to

Question: _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569427

Topic Name: Mathematics-Section B

Question: Let $a_n = \int_{-1}^{n} \left(1 + \frac{x}{2} + \frac{x^2}{3} + \ldots\ldots + \frac{x^{n-1}}{n}\right) dx$ for every $n \in \mathbb{N}$. Then the sum of all the elements of the set $\{n \in \mathbb{N} : a_n \in (2, 30)\}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569428
Topic Name: Mathematics-Section B

Question: If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2\left(3 - \sqrt{3}\right)x + 2\left(4 - \sqrt{6}\right)y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$, touch internally at the point $P(\alpha, \beta)$, then $\left(\alpha + \sqrt{3}\right)^2 + \left(\beta + \sqrt{6}\right)^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569429
Topic Name: Mathematics-Section B

Question: Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$ at the point $(-2, 3)$ be A. Then $8A$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569430
Topic Name: Mathematics-Section B

Question: Let $x = \sin(2\tan^{-1}\alpha)$ and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \{\alpha \in \mathbb{R} : y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569431
Topic Name: Physics-Section A

Question: In AM modulation, a signal is modulated on a carrier wave such that maximum and minimum amplitudes are found to be 6 V and 2 V respectively. The modulation index is :

A 100%

B 80%

C 60%

D 50%

Answer Given By Candidate:**D**

Question ID:1569432
Topic Name: Physics-Section A

Question: The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is

A $\frac{5}{2}$

B $\frac{25}{4}$

C $\frac{5}{4}$

D $\frac{25}{2}$

Answer Given By Candidate:**B**

Question ID:1569433
Topic Name: Physics-Section A

Question: A drop of liquid of density **ρ** is floating half immersed in a liquid of density **σ** and surface tension 7.5×10^{-4} Ncm^{-1}. The radius of drop in cm will be : $(g = 10ms^{-2})$

A $\frac{15}{\sqrt{(2\rho - \sigma)}}$

B $\frac{15}{\sqrt{(\rho - \sigma)}}$

C $\frac{3}{2\sqrt{(\rho - \sigma)}}$

D $\frac{3}{20\sqrt{(2\rho - \sigma)}}$

Answer Given By Candidate:**A**

Question ID:1569434
Topic Name: Physics-Section A

Question: Two billiard balls of mass 0.05 kg each moving in opposite directions with $10\ ms^{-1}$ collide and rebound with the same speed. If the time duration of contact is $t = 0.005$ s, then what is the force exerted on the ball due to each other ?

A 100 N

B 200 N

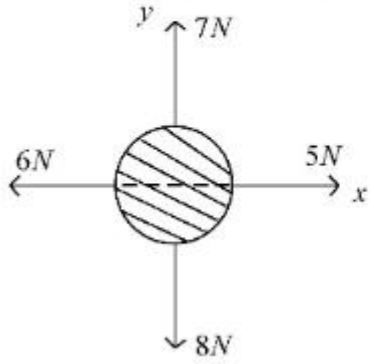
C 300 N

D 400 N

Answer Given By Candidate:**B**

Question ID:1569435
Topic Name: Physics-Section A

Question: For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero ?



A $\sqrt{2}N, 45^\circ$

B $\sqrt{2}N, 135^\circ$

C $\frac{2}{\sqrt{3}}N, 30^\circ$

D $2N, 45^\circ$

Answer Given By Candidate:**A**

Question ID:1569436
Topic Name: Physics-Section A

Question: Capacitance of an isolated conducting sphere of radius $\mathbf{R_1}$ becomes n times when it is enclosed by a concentric conducting sphere of radius $\mathbf{R_2}$ connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is :

A

$\frac{n}{n-1}$

B $\frac{2n}{2n+1}$

C $\frac{n+1}{n}$

D $\frac{2n+1}{n}$

Answer Given By Candidate:**C**

Question ID:1569437
Topic Name: Physics-Section A

Question: The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is $1:\sqrt{2}$. Then, the ratio of V_p to V_d will be :

A 1:1

B $\sqrt{2}:1$

C 2 : 1

D 4 : 1

Answer Given By Candidate:**D**

Question ID:1569438
Topic Name: Physics-Section A

Question: For an object placed at a distance 2.4 m from a lens, a sharp focused image is observed on a screen placed at a distance 12 cm form the lens. A glass plate of refractive index 1.5 and thickness 1 cm is introduced between lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screeen ?

A 0.8 m

B 3.2 m

C 1.2 m

D 5.6 m

Answer Given By Candidate:**Not Attempted**

Question ID:1569439
Topic Name: Physics-Section A

Question: Light wave traveling in air along x-direction is given by $E_y = 540 \text{ Sin } \pi \times 10^4(x - ct) Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 \text{ ms}^{-1}$)

A $18 \times 10^{-7}\ T$

B $54 \times 10^{-7}\ T$

C $54 \times 10^{-8}\ T$

D $18 \times 10^{-8}\ T$

Answer Given By Candidate:**A**

Question ID:1569440
Topic Name: Physics-Section A

Question: When you walk through a metal detector carrying a metal object in your pocket, it raises an alarm. This phenomenon works on :

A Electromagnetic induction

B Resonance in ac circuits

C Mutual induction in ac circuits

D Interference of electromagnetic waves

Answer Given By Candidate:**B**

Question ID:1569441
Topic Name: Physics-Section A

Question: An electron with energy 0.1 keV moves at right angle to the earth's magnetic field of 1×10^{-4} Wbm^{-2}. The frequency of revolution of the electron will be (Take mass of electron $= 9.0 \times 10^{-31}$ kg)

A 1.6×10^5 Hz

B 5.6×10^5 Hz

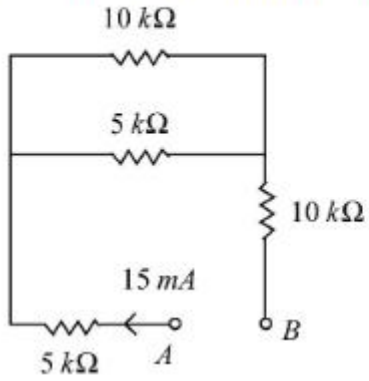
C 2.8×10^6 Hz

D 1.8×10^6 Hz

Answer Given By Candidate:**C**

Question ID:1569442
Topic Name: Physics-Section A

A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be :



Question:

A 50 V

B 75 V

C 150 V

D 275 V

Answer Given By Candidate:**D**

Question ID:1569443
Topic Name: Physics-Section A

The length of a seconds pendulum at a height h = 2R from earth surface will be :

(Given R = Radius of earth and acceleration due to gravity at the surface of earth, $g = \pi^2\ ms^{-2}$)

Question:

A $\frac{2}{9}$ m

B $\frac{4}{9}$ m

C $\frac{8}{9}$ m

D $\frac{1}{9}$ m

Answer Given By Candidate:**D**

Question ID:1569444
Topic Name: Physics-Section A

Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the

Question: value of n will be :

A 1

B 2

C 3

D 4

Answer Given By Candidate:**Not Attempted**

Question ID:1569445
Topic Name: Physics-Section A

Question: Let η_1 is the efficiency of an engine at $T_1 = 447°C$ and $T_2 = 147°C$ while η_2 is the efficiency at $T_1 = 947°C$ and $T_2 = 47°C$ The ratio $\frac{\eta_1}{\eta_2}$ will be :

A 0.41

B 0.56

C 0.73

D 0.70

Answer Given By Candidate:**B**

Question ID:1569446
Topic Name: Physics-Section A

Question: An object is taken to a height above the surface of earth at a distance $\frac{5}{4}$ R from the centre of the earth. Where radius of earth, R = 6400 km. The percentage decrease in the weight of the object will be :

A 36%

B 50%

C 64%

D 25%

Answer Given By Candidate:**A**

Question ID:1569447
Topic Name: Physics-Section A

Question: A bag of sand of mass 9.8 kg is suspended by a rope. A bullet of 200 g travelling with speed 10 ms^{-1} gets embedded in it, then loss of kinetic energy will be :

A 4.9 J

B 9.8 J

C 14.7 J

D 19.6 J

Answer Given By Candidate:**B**

Question ID:1569448
Topic Name: Physics-Section A

Question: A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then 'tan θ' will be equal to :

A $\frac{1}{4}$

B $\frac{1}{2}$

C 2

D 4

Answer Given By Candidate:**D**

Question ID:1569449
Topic Name: Physics-Section A

Question: The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :

A 2

B 4

C 6

D 8

Answer Given By Candidate:**D**

Question ID:1569450
Topic Name: Physics-Section A

Question: Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength λ . The value of principal quantum number 'n' of the excited state will be : (R : Rydberg constant)

A $\sqrt{\frac{\lambda R}{\lambda - 1}}$

B $\sqrt{\frac{\lambda R}{\lambda R - 1}}$

C $\sqrt{\frac{\lambda}{\lambda R - 1}}$

D $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

Answer Given By Candidate:**B**

Question ID:1569451
Topic Name: Physics-Section B

Question: A particle is moving in a straight line such that its velocity is increasing at 5 ms^{-1} per meter. The acceleration of the particle is _____ ms^{-2} at a point where its velocity is 20 ms^{-1}.

Answer Given By Candidate:**Not Attempted**

Question ID:1569452
Topic Name: Physics-Section B

Question: Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3 m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}$ m. The value of x is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569453
Topic Name: Physics-Section B

Question: A block of ice of mass 120 g at temperature 0°C is put in 300 g of water at 25°C. The xg of ice melts as the temperature of the water reaches 0°C. The value of x is _____.
[Use specific heat capacity of water = 4200 $Jkg^{-1}K^{-1}$, Latent heat of ice = 3.5×10^5 Jkg^{-1}]

Answer Given By Candidate:**Not Attempted**

Question ID:1569454
Topic Name: Physics-Section B

Question: $\frac{x}{x+4}$ is the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its

(i) third permitted energy level to the second level and

(ii) the highest permitted energy level to the second permitted level.

The value of x will be _____.

Answer Given By Candidate:**5**

Question ID:1569455
Topic Name: Physics-Section B

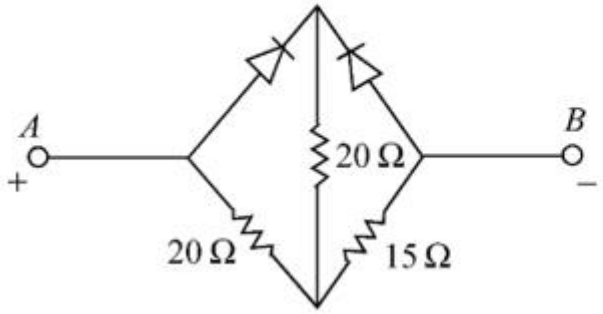
In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 36 cm length of wire. This cell is now replaced by another cell of emf 1.80 V. The difference in balancing length of potentiometer wire in above conditions will be

Question: _____ cm.

Answer Given By Candidate:**18**

Question ID:1569456
Topic Name: Physics-Section B

Two ideal diodes are connected in the network as shown is figure. The equivalent resistance between A and B is _____ Ω .



Question:

Answer Given By Candidate:**25**

Question ID:1569457
Topic Name: Physics-Section B

Two waves executing simple harmonic motions travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is

Question: _____ (degree).

Answer Given By Candidate:**60**

Question ID:1569458
Topic Name: Physics-Section B

Question: Two parallel plate capacitors of capacity C and 3C are connected in parallel combination and charged to a potential difference 18 V. The battery is then disconnected and the space between the plates of the capacitor of capacity C is completely filled with a material of dielectric constant 9. The final potential difference across the combination of capacitors will be _____V.

Answer Given By Candidate:**Not Attempted**

Question ID:1569459
Topic Name: Physics-Section B

A convex lens of focal length 20 cm is placed in front of a convex mirror with principal axis coinciding each other. The distance between the lens and mirror is 10 cm. A point object is placed on principal axis at a distance of 60 cm from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is

Question: _____ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:1569460
Topic Name: Physics-Section B

Magnetic flux (in weber) in a closed circuit of resistance 20 Ω varies with time t(s) as $\phi = 8t^2 - 9t + 5$. The magnitude of the induced current at t = 0.25 s will be

Question: _____ mA.

Answer Given By Candidate:**250**

Question ID:1569461
Topic Name: Chemistry-Section A

Match **List I** with **List II** :

List I **(molecule)**	**List II** **(hybridization ; shape)**
A. XeO_3	I. sp^3d ; linear
B. XeF_2	II. sp^3; pyramidal
C. $XeOF_4$	III. sp^3d^3; distorted octahedral
D. XeF_6	IV. sp^3d^2; square pyramidal

Question: Choose the correct answer from the options given below:

A A-II, B-I, C-IV, D-III

B A-II, B-IV, C-III, D-I

C A-IV, B-II, C-III, D-I

D A-IV, B-II, C-I, D-III

Answer Given By Candidate:**A**

Question ID:1569462

Topic Name: Chemistry-Section A

Question: Two solutions A and B are prepared by dissolving 1 g of non-volatile solutes X and Y, respectively in 1 kg of water. The ratio of depression in freezing points for A and B is found to be 1:4. The ratio of molar masses of X and Y is

A 1:4

B 1:0.25

C 1:0.20

D 1:5

Answer Given By Candidate:**B**

Question ID:1569463

Topic Name: Chemistry-Section A

Question: K_{a_1}, K_{a_2} and K_{a_3} are the respective ionization constants for the following reactions (a), (b), and (c).

(a) $H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$

(b) $HC_2O_4^- \rightleftharpoons H^+ + HC_2O_4^{2-}$

(c) $H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-}$

The relationship between K_{a_1}, K_{a_2} and K_{a_3} is given as

A $K_{a_3} = K_{a_1} + K_{a_2}$

B $K_{a_3} = K_{a_1} - K_{a_2}$

C $K_{a_3} = K_{a_1} / K_{a_2}$

D $K_{a_3} = K_{a_1} \times K_{a_2}$

Answer Given By Candidate:**D**

Question ID:1569464

Topic Name: Chemistry-Section A

Question: The molar conductivity of a conductivity cell filled with 10 moles of 20 mL NaCl solution is Λ_{m1} and that of 20 moles another identical cell heaving 80 mL NaCl solution is Λ_{m2}. The conductivities exhibited by these two cells are same. The relationship between Λ_{m2} and Λ_{m1} is

A $\Lambda_{m2} = 2\Lambda_{m1}$

B $\Lambda_{m2} = \Lambda_{m1} / 2$

C

$\Lambda_{m2} = \Lambda_{m1}$

D $\Lambda_{m2} = 4\Lambda_{m1}$

Answer Given By Candidate:**A**

Question ID:1569465
Topic Name: Chemistry-Section A
Question:
For micelle formation, which of the following statements are correct ?

A. Micelle formation is an exothermic process.
B. Micelle formation is an endothermic process.
C. The entropy change is positive.
D. The entropy change is negative.

A A and D only

B A and C only

C B and C only

D B and D only

Answer Given By Candidate:**A**

Question ID:1569466
Topic Name: Chemistry-Section A
Question:
The first ionization enthalpies of Be, B, N and O follow the order

A O < N < B < Be

B Be < B < N < O

C B < Be < N < O

D B < Be < O < N

Answer Given By Candidate:**D**

Question ID:1569467
Topic Name: Chemistry-Section A
Question:
Given below are two statements.
Statement I: Pig iron is obtained by heating cast iron with scrap iron.
Statement II: Pig iron has a relatively lower carbon content than that of cast iron.
In the light of the above statements, choose the *correct* answer from the options given below.

A Both Statement I and Statement II are correct.

B Both Statement I and Statement II are **not** correct.

C Statement I is correct but Statement II is **not** correct.

D Statement I is **not** correct but Statement II is correct.

Answer Given By Candidate:**D**

Question ID:1569468
Topic Name: Chemistry-Section A
Question: High purity (>99.95%) dihydrogen is obtained by

A reaction of zinc with aqueous alkali.

B electrolysis of acidified water using platinum electrodes.

C electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

D reaction of zinc with dilute acid.

Answer Given By Candidate:**C**

Question ID:1569469
Topic Name: Chemistry-Section A
Question: The correct order of density is

A Be > Mg > Ca > Sr

B Sr > Ca > Mg > Be

C Sr > Be >Mg > Ca

D Be > Sr > Mg > Ca

Answer Given By Candidate:**D**

Question ID:1569470
Topic Name: Chemistry-Section A
Question: The total number of acidic oxides from the following list is
$NO, N_2O, B_2O_3, N_2O_5, CO, SO_3, P_4O_{10}$

A 3

B 4

C 5

D 6

Answer Given By Candidate:**B**

Question ID:1569471
Topic Name: Chemistry-Section A

Question: The correct order of energy of absorption for the following metal complexes is
A : $[Ni(en)_3]^{2+}$, B : $[Ni(NH_3)_6]^{2+}$, C : $[Ni(H_2O)_6]^{2+}$

A C < B < A

B B < C < A

C C < A < B

D A < C < B

Answer Given By Candidate:**A**

Question ID:1569472
Topic Name: Chemistry-Section A

Question: Match **List I** with **List II**.

List I	List II
A. Sulphate	I. Pesticide
B. Fluoride	II. Bending of bones
C. Nicotine	III. Laxative effect
D. Sodium arsinite	IV. Herbicide

Choose the correct answer from the options given below:

A A-II, B-III, C-IV, D-I

B A-IV, B-III, C-II, D-I

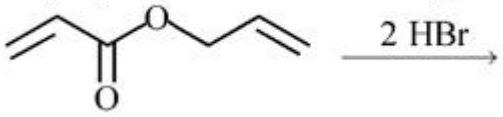
C A-III, B-II, C-I, D-IV

D A-III, B-II, C-IV, D-I

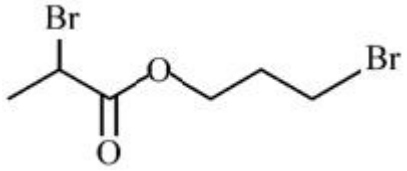
Answer Given By Candidate:**C**

Question ID:1569473
Topic Name: Chemistry-Section A

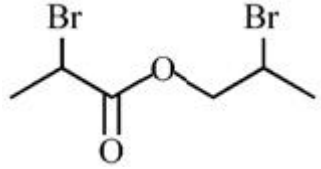
Question: Major product of the following reaction is



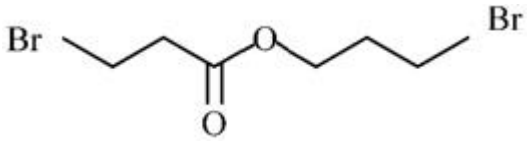
A



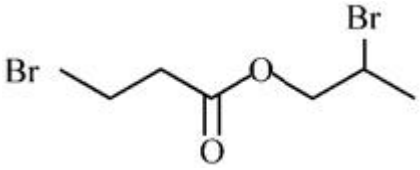
B



C



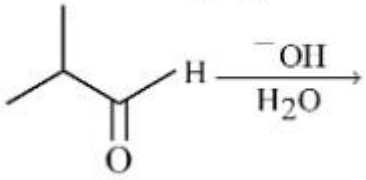
D



Answer Given By Candidate:**D**

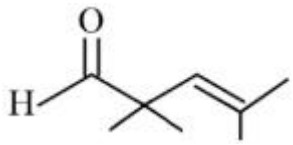
Question ID:1569474
Topic Name: Chemistry-Section A

What is the major product of the following reaction ?

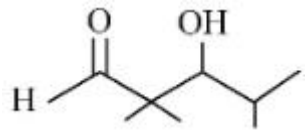


Question:

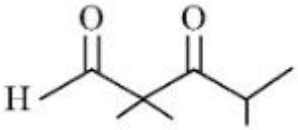
A



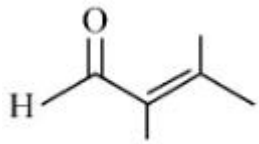
B



C



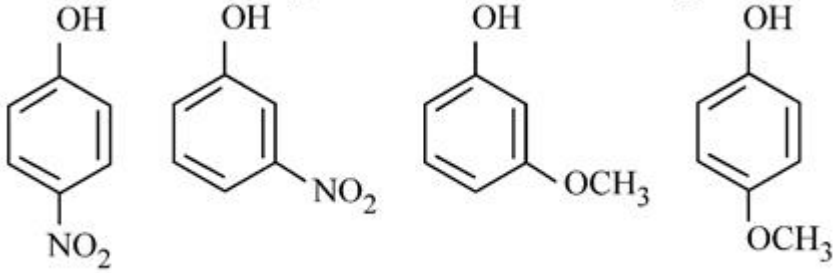
D



Answer Given By Candidate:**B**

Question ID:1569475
Topic Name: Chemistry-Section A

Arrange the following in decreasing acidic strength.



(A) (B) (C) (D)

Question:

A A > B > C > D

B B > A > C > D

C D > C > A > B

D D > C > B > A

Answer Given By Candidate:**A**

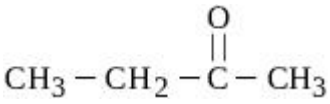
Question ID:1569476
Topic Name: Chemistry-Section A

$$CH_3-CH_2-CN \xrightarrow[\text{Ether}]{CH_3MgBr} A \xrightarrow{H_3O^+} B \xrightarrow[HCl]{Zn-Hg} C$$

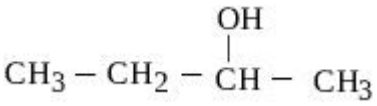
Question: The correct structure of **C** is

A $CH_3-CH_2-CH_2-CH_3$

B



C



D $CH_3-CH_2-CH=CH_2$

Answer Given By Candidate:**A**

Question ID:1569477
Topic Name: Chemistry-Section A

Match **List I** with **List II** :

List I Polymer	List II used for items
A. Nylon 6,6	I. Buckets
B. Low density polythene	II. Non-stick utensils
C. High density polythene	III. Bristles of brushes
D. Teflon	IV. Toys

Question: Choose the correct answer from the options given below:

A A-III, B-I, C-IV, D-II

B A-III, B-IV, C-I, D-II

C A-II, B-I, C-IV, D-III

D A-II, B-IV, C-I, D-III

Answer Given By Candidate:**B**

Question ID:1569478
Topic Name: Chemistry-Section A
Question: Glycosidic linkage between C1 of α-glucose and C2 of β–fructose is found in

A maltose

B sucrose

C lactose

D amylose

Answer Given By Candidate:**B**

Question ID:1569479
Topic Name: Chemistry-Section A
Question: Some drugs bind to a site other than the active site of an enzyme. This site is known as

A non-active site

B allosteric site

C competitive site

D therapeutic site

Answer Given By Candidate:**B**

Question ID:1569480
Topic Name: Chemistry-Section A
Question: In base vs. acid titration, at the end point methyl orange is present as

A quinonoid form

B heterocyclic form

C phenolic form

D benzenoid form

Answer Given By Candidate:**A**

Question ID:1569481
Topic Name: Chemistry-Section B
Question:56.0 L of nitrogen gas is mixed with excess of hydrogen gas and it is found that 20 L of ammonia gas is produced. The volume of unused nitrogen gas is found to be __ L.

Answer Given By Candidate:**2**

Question ID:1569482
Topic Name: Chemistry-Section B
Question:A sealed flask with a capacity of 2 dm^3 contains 11 g of propane gas. The flask is so weak that it will burst if the pressure becomes 2 MPa. The minimum temperature at which the flask will burst is _____ °C. [Nearest integer] (Given: R = 8.3 $J\ K^{-1}\ mol^{-1}$, Atomic masses of C and H are 12u and 1u, respectively.)(Assume that propane behaves as an ideal gas.)

Answer Given By Candidate:**Not Attempted**

Question ID:1569483
Topic Name: Chemistry-Section B
Question: When the excited electron of a H atom from n = 5 drops to the ground state, the maximum number of emission lines observed are _____.

Answer Given By Candidate:**4**

Question ID:1569484
Topic Name: Chemistry-Section B

While performing a thermodynamics experiment, a student made the following observations.

$HCl + NaOH \rightarrow NaCl + H_2O \ \Delta H = -57.3 \text{ kJ mol}^{-1}$

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O \ \Delta H = -55.3 \text{ kJ mol}^{-1}$

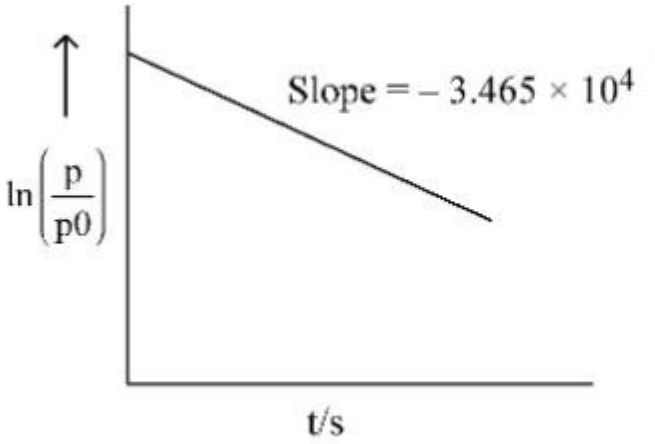
The enthalpy of ionization of CH_3COOH as calculated by the student is

Question: _____ kJ mol^{-1}. (nearest integer)

Answer Given By Candidate:**2**

Question ID:1569485
Topic Name: Chemistry-Section B

For the decomposition of azomethane,
$CH_3N_2CH_3(g) \rightarrow CH_3CH_3(g) + N_2(g)$, a first order reaction, the variation in partial pressure with time at 600 K is given as



Question: The half life of the reaction is ___ $\times 10^{-5}$s. [Nearest integer]

Answer Given By Candidate:**Not Attempted**

Question ID:1569486
Topic Name: Chemistry-Section B

Question: The sum of number of lone pairs of electrons present on the central atoms of XeO_3, $XeOF_4$ and XeF_6, is _____

Answer Given By Candidate:**3**

Question ID:1569487
Topic Name: Chemistry-Section B

Question: The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs Cr^{3+}/Cr^{2+}, Mn^{3+}/Mn^{2+}, Fe^{3+}/Fe^{2+} and Co^{3+}/Co^{2+} that has negative standard electrode potential, is ___ B.M. [Nearest integer]

Answer Given By Candidate:**Not Attempted**

Question ID:1569488
Topic Name: Chemistry-Section B
Question: A sample of 4.5 mg of an unknown monohydric alcohol, R-OH was added to methylmagnesium iodide. A gas is evolved and is collected and its volume measured to be 3.1 mL. The molecular weight of the unknown alcohol is __ g/mol. [Nearest integer]

Answer Given By Candidate:**Not Attempted**

Question ID:1569489
Topic Name: Chemistry-Section B
Question: The separation of two coloured substances was done by paper chromatography. The distances travelled by solvent front, substance A and substance B from the base line are 3.25 cm, 2.08 cm and 1.05 cm, respectively. The ratio of R_f values of A to B is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1569490
Topic Name: Chemistry-Section B
Question: The total number of monobromo derivatives formed by the alkanes with molecular formula C_5H_{12} is (excluding stereo isomers) _____.

Answer Given By Candidate:**8**